

Day 2: HTML Basics — Complete Teacher's Notes

Fully detailed lecture notes with explanations, examples, analogies, tables & teaching tips

1. HTML Introduction

HTML = HyperText Markup Language.

- **HyperText** — text containing links that connect to other text/pages.
- **Markup Language** — uses tags to label/wrap content so the browser knows how to display it.

HTML is **not a programming language** — it has no logic, loops, or calculations. It only describes *structure*: what each piece of content *is* (heading, paragraph, list, image) — not how it looks (CSS) or how it behaves (JavaScript).

Teaching analogy: Compare HTML to the skeleton of a body — bones give structure, skin (CSS) gives appearance, muscles/nerves (JavaScript) give movement/behavior. Very effective for beginners.

1.1 How the Browser Reads HTML

When you open an HTML file, the browser reads the code top to bottom, builds something called the **DOM (Document Object Model)** — a tree-like structure of all elements — and then paints it on screen. This is why nesting and closing tags matter: a broken tree can cause the whole page to render incorrectly.

1.2 History in Brief

HTML was created by Tim Berners-Lee in 1991. It has evolved through versions (HTML 2, 3.2, 4.01, XHTML) up to the current standard, **HTML5**, which added semantic tags, audio/video support, and better mobile compatibility.

1.3 Basic Tag Syntax

```
<tagname attribute="value">content</tagname>
```

Self-closing / void tags (no closing tag, no content inside):

```
<br>

<hr>
<input type="text">
<meta charset="UTF-8">
```

Common mistake: students often try to write `<br></br>` or `<img></img>`. Void tags never take a closing tag.

1.4 Block vs Inline Elements

Every HTML element falls into one of two display categories:

Type	Behavior	Examples
Block-level	Starts on a new line, takes full available width	<h1>-<h6>, <p>, <div>, , , , <table>, <form>
Inline	Stays in the flow of text, only takes as much width as needed	<a>, , , , <i>, ,

Teaching tip: Draw two boxes on the whiteboard — one stretching full width (block) and one hugging its content (inline) — students remember this visually.

2. HTML Document Structure

```
<!DOCTYPE html>
<html>
<head>
  <title>My Web Page</title>
</head>
<body>
  <h1>Hello World</h1>
  <p>This is my first web page.</p>
</body>
```

</html>

Part	Meaning
<!DOCTYPE html>	Declares HTML5 document — must be the very first line.
<html>	Root element — everything else is nested inside. Can take a <code>lang</code> attribute, e.g. <code>lang="en"</code> .
<head>	Metadata — not visible on page (title, charset, linked CSS/JS files, favicon, SEO tags).
<title>	Text shown on the browser tab and used by search engines.
<body>	Everything visible to the user.

2.1 Common <head> Additions

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<meta name="description" content="Short description of the page for search engines">
<link rel="stylesheet" href="style.css">
<link rel="icon" href="favicon.ico">
<script src="script.js"></script>
```

Explain to students: UTF-8 supports almost every character/emoji worldwide; the viewport meta tag makes the page mobile-responsive; the meta description affects how the page appears in Google search results.

2.2 Whitespace & Indentation

HTML ignores extra whitespace, tabs, and blank lines in the source code — but developers still indent code consistently (usually 2 or 4 spaces per nested level) purely for **readability**, not because the browser needs it.

3. Elements and Attributes

Element = opening tag + content + closing tag.

```
<p>This is a paragraph.</p>
```

Attribute = extra info added inside the opening tag as `name="value"` pairs. An element can have multiple attributes, separated by spaces.

```
<a href="https://google.com" target="_blank" title="Search engine">Visit Google</a>
```

Attribute	Used on	Purpose
href	<a>	Link destination
src	, <script>	File path/URL of the resource
alt		Fallback text / accessibility
id	any	Unique identifier for one element (used only once per page)
class	any	Group identifier (used in CSS/JS, reusable across many elements)
style	any	Inline CSS styling
title	any	Tooltip on hover
target	<a>	Where to open the link (<code>_blank</code> = new tab)
width / height	, <video>	Dimensions in pixels
placeholder	<input>	Hint text shown inside an empty input field
disabled	form elements	Disables/greys out the element

3.1 id vs class — Important Distinction

Feature	id	class
Uniqueness	Must be unique — only one element per page	Can be reused on many elements
CSS selector	#myId	.myClass

Use case	Targeting one specific element (e.g. a single header)	Styling a group of similar elements (e.g. all buttons)
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Common student mistake: forgetting quotes around attribute values, or putting attributes in the closing tag. Emphasize: attributes exist **ONLY** in the opening tag.

4. Headings

```
<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>
```

- Only **one <h1>** per page (main title).
- Don't skip levels just to change size — use CSS for styling; headings exist for structure/SEO/accessibility, not appearance.
- Screen readers let visually impaired users "jump" between headings — poor heading structure breaks this navigation.

Teaching tip: Compare headings to a book's table of contents — h1 = book title, h2 = chapter, h3 = sub-section, h4-h6 = deeper sub-points.

5. Paragraphs

```
<p>This is paragraph one.</p>
<p>This is paragraph two.</p>
```

- Browser automatically adds space before/after each paragraph.
- Extra spaces/line breaks typed in code are **collapsed** into a single space by the browser (this is called "whitespace collapsing").
- Use `<br>` for a forced line break within the same block of text (not for spacing between paragraphs).

```
<p>Roses are red <br> Violets are blue</p>
```

5.1 Basic Text Formatting Tags (often taught alongside paragraphs)

Tag	Effect
<code>text</code>	Bold (visual only)
<code>text</code>	Bold + tells screen readers it's important
<code><i>text</i></code>	<i>Italic</i> (visual only)
<code>text</code>	<i>Italic</i> + adds emphasis meaning
<code><u>text</u></code>	Underline
<code><mark>text</mark></code>	Highlighted text
<code><small>text</small></code>	Smaller text
<code>text</code>	Strikethrough (deleted text)
<code><sub>text</sub></code>	Subscript (e.g. H ₂ O)
<code><sup>text</sup></code>	Superscript (e.g. x ²)

Explain the difference between `<b>/<strong>` and `<i>/<em>`: visually identical, but `<strong>` and `<em>` carry semantic meaning that assistive technology understands.

6. Lists

6.1 Unordered List

```
<ul>
  <li>Apple</li>
```

```
<li>Banana</li>
</ul>
```

6.2 Ordered List

```
<ol>
  <li>Wake up</li>
  <li>Brush teeth</li>
</ol>
```

Attributes: `type="A"` (A,B,C), `type="a"` (a,b,c), `type="i"` (i,ii,iii), `type="I"` (I,II,III), `start="5"` (start numbering from 5), `reversed` (counts backward).

6.3 Description List

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

Rule to stress: `<li>` must always live inside `<ul>` or `<ol>` — never standalone. Similarly, `<dt>/<dd>` must live inside `<dl>`.

6.4 Nested Lists

```
<ul>
  <li>Fruits
    <ul>
      <li>Apple</li>
      <li>Mango</li>
    </ul>
  </li>
  <li>Vegetables</li>
</ul>
```

Nesting is used for menus, sitemaps, folder structures, and multi-level navigation bars.

7. Links

```
<a href="https://www.google.com">Go to Google</a>
```

Type	Example
External	Wikipedia
Internal	About
Same-page jump	Go to Contact + <h2 id="contact">
Email	Email
Phone	Call
Download	Download PDF

Useful attributes: `target="_blank"` (new tab), `title="..."` (tooltip), `rel="noopener noreferrer"` (security best-practice when using `target="_blank"`).

Teaching tip: Show students what happens if you forget the "https://" in an external href — the browser treats it as a broken internal link. Live demo this mistake.

7.1 Relative vs Absolute Paths

Path Type	Example	Meaning
Absolute	https://example.com/about.html	Full web address — works from anywhere
Relative	about.html	Same folder as current page

Relative (subfolder)	pages/about.html	Inside a subfolder
Relative (parent)	../index.html	Go up one folder level

8. Images

```

```

Attribute	Purpose
src	Path/URL of image (required)
alt	Alt text — accessibility & fallback (important!)
width/height	Size in pixels — helps prevent layout shift while loading
title	Tooltip on hover
loading="lazy"	Delays loading offscreen images for better performance

Stress that `alt` is not optional in professional practice — screen readers depend on it, and search engines use it for image indexing.

8.1 Supported Image Formats

Format	Best For
.jpg / .jpeg	Photographs (smaller file size, some quality loss)
.png	Graphics/logos needing transparency
.gif	Simple animations
.svg	Vector icons/logos (scales without quality loss)
.webp	Modern format — smaller size, good quality

8.2 Image as a Link

Images can also be wrapped inside an anchor tag to become clickable:

```
<a href="https://google.com">
  
</a>
```

9. Comments

```
<!-- This is a comment -->
<!--
  Multi-line
  comment
-->
```

Comments are invisible on the rendered page. Uses:

- Leaving notes for yourself or your team (e.g. "TODO: fix later")
- Temporarily disabling a piece of code without deleting it, while testing
- Marking the start/end of major sections in a large file (e.g. `<!-- Header Section -->`, `<!-- Footer Section -->`)

Comments are visible in "View Page Source" — never put sensitive information (passwords, secret keys) inside HTML comments.

10. HTML Entities

Some characters have special meaning in HTML syntax itself (like `<`, `>`, `&`), so typing them directly can confuse the browser. Entities let you display these characters safely as plain text.

Character	Entity Name	Entity Number
<	<	<

>	>	>
&	&	&
"	"	"
'	'	'
(space)	 	
©	©	©
®	®	®
™	™	™
€	€	€
₹	—	₹
×	×	×
÷	÷	÷
→	→	→
♥	♥	♥

Teaching tip: Demonstrate live — type `<b>` as literal text in a paragraph without entities, show the browser breaks it, then fix using `<b>`.

11. Introduction to Semantic HTML (Bonus Topic)

Semantic tags describe the *meaning* of content, not just its appearance. HTML5 introduced several semantic tags that are commonly used alongside the basics above:

Tag	Purpose
<header>	Top section — logo, navigation, title
<nav>	Navigation links/menu
<main>	Main unique content of the page
<section>	A thematic grouping of content
<article>	Self-contained content (e.g. blog post)
<aside>	Side content — related links, ads
<footer>	Bottom section — copyright, contact info

These aren't required for Day 2, but mentioning them early helps students recognize them in future lessons and understand why "just use `<div>` for everything" is discouraged in modern HTML.

12. Common Beginner Mistakes — Quick Checklist

Mistake	Correction
Forgetting closing tags	Every non-void tag needs a closing tag: <code><p>...</p></code>
Wrong nesting order	Close tags in reverse order they were opened: <code><i>text</i></code>
Missing quotes on attributes	Always use <code>href="..."</code> not <code>href=...</code>
Using multiple <code><h1></code> tags	Only one <code><h1></code> per page
Broken image paths	Double-check <code>src</code> path/spelling and file extension
Typing raw <code><</code> or <code>></code> in text	Use <code>&lt;</code> and <code>&gt;</code> entities instead
Putting <code></code> outside <code>/</code>	Always wrap list items properly

Practical: Personal Profile Page (Teacher Walkthrough)

Live-code this with students step by step. Suggested build order:

1. Set up boilerplate (doctype, head, meta, title)
2. Add h1/h2 name & role
3. Add profile image with alt text
4. Add About Me paragraph (use & entity example inside it)
5. Add Hobbies (unordered list) and Skills (ordered list)
6. Add Contact section with id for anchor demo, mailto and tel links
7. Add social links with target="_blank"
8. Add a comment (TODO note) to show real-world usage
9. Finish with <hr> and copyright line using ©

Assessment idea: After the walkthrough, ask students to independently rebuild the page with their own details — this reinforces retention better than copy-paste.

13. Recap Summary Table

Topic	Key Tag(s)
Structure	<html>, <head>, <body>
Headings	<h1> - <h6>
Paragraph	<p>,
Lists	, , , <dl>
Links	
Images	
Comments	<!-- -->
Entities	< > & ©

14. Void (Self-Closing) Elements — Full Reference

Besides
, , and <hr>, HTML has several other void elements students will meet soon:

Tag	Purpose
 	Line break
<hr>	Horizontal rule / divider line
	Embed an image
<input>	Form input field
<meta>	Metadata in <head>
<link>	Link external resource (CSS, favicon)
<source>	Media source for <video>/<audio>
<area>	Clickable region in an image map
<col>	Column properties inside a table

Give students a simple rule: "If the tag can never wrap content, it's void — no closing tag."

15. Case Sensitivity & Writing Style Conventions

- HTML tag names are **not case-sensitive** — <P> and <p> work the same — but lowercase is the universal industry standard.
- Attribute names are also not case-sensitive, but again, lowercase is standard practice.
- File names (like image paths) ARE case-sensitive on many servers — "Photo.JPG" ≠ "photo.jpg".

Teach students to always write tags and attributes in lowercase from day one — it avoids many hard-to-spot bugs later.

16. Validating HTML

Professional developers check their code using the **W3C Markup Validator** (validator.w3.org) to catch structural errors — unclosed tags, missing attributes, incorrect nesting — before publishing a site. Encourage students to validate their practical assignment once it's complete.

17. Mini Glossary

Term	Meaning
Tag	Keyword wrapped in angle brackets, e.g. <p>
Element	A tag plus its content and closing tag
Attribute	Extra property added to a tag's opening tag
DOM	Tree structure the browser builds from HTML to render/manipulate the page
Void element	A tag with no closing tag and no content
Semantic tag	A tag that describes the meaning of its content, not just appearance
Nesting	Placing one element inside another
Whitespace collapsing	Browser's habit of reducing multiple spaces/line breaks to one space

18. Classroom Activity Ideas

- Tag Match-Up:** Give students a shuffled list of tags and their purposes; ask them to match each tag to its correct use.
- Spot the Bug:** Show a broken HTML snippet (missing closing tag, wrong nesting) on the screen and ask students to identify and fix the error.
- Speed Build:** In pairs, students race to build a 3-item list with a heading and one image in under 5 minutes.
- Entity Hunt:** Give a sentence containing symbols (<, >, &, ©) and ask students to rewrite it safely using entities.

19. Frequently Asked Questions (For Class Discussion)

Q: Can I skip the DOCTYPE declaration?

Technically the page may still render, but the browser may switch to "quirks mode," causing inconsistent rendering across browsers. Always include it.

Q: Do I need to close tags?

Browsers are forgiving and often auto-correct this, but professional code always explicitly closes every non-void tag.

Q: What happens if two elements have the same id?

The HTML will still render, but it becomes invalid and can break CSS/JavaScript that targets that id, since only the first match is typically used.

Q: Why use instead of just styling text bold with CSS?

 carries semantic meaning for screen readers and search engines — CSS bold is purely visual with no meaning attached.

20. Extra Practice Snippets (For Board Work)

Snippet A — Simple table of contents using links:

```
<h2>Table of Contents</h2>
<ul>
  <li><a href="#intro">Introduction</a></li>
  <li><a href="#about">About</a></li>
</ul>
<h3 id="intro">Introduction</h3>
```

```
<p>...</p>
<h3 id="about">About</h3>
<p>...</p>
```

Snippet B — Recipe card using an ordered list:

```
<h2>Tea Recipe</h2>
<ol>
  <li>Boil water</li>
  <li>Add tea leaves</li>
  <li>Add milk & sugar</li>
  <li>Strain and serve</li>
</ol>
```

Snippet C — Simple image gallery row:

```
<h2>Gallery</h2>



```

21. Recap Summary Table

Topic	Key Tag(s)
Structure	<html>, <head>, <body>
Headings	<h1> - <h6>
Paragraph	<p>,
Text formatting	, , <i>, , <mark>
Lists	, , , <dl>
Links	
Images	
Comments	<!-- -->
Entities	< > & ©
Semantic preview	<header>, <nav>, <main>, <footer>

End-of-class checklist for the teacher: Have all students successfully opened their profile.html in a browser? Did each student's page include at least one list, one link, one image, and one comment? Address common errors as a group before moving to Day 3.

22. Homework Assignment

- Assign the following for students to complete before the next class:
1. Finish and polish the Personal Profile Page started in class.
 2. Add a fourth section titled "My Favorite Quote" using a paragraph tag.
 3. Add a nested list under Hobbies with at least one sub-item.
 4. Validate the finished page using the W3C Validator and fix any reported errors.
 5. Take a screenshot of the rendered page and submit it along with the .html file.

Grading suggestion: Award marks separately for structure correctness, proper use of each tag type covered today, and creativity/content quality.

23. Looking Ahead — Day 3 Preview

- Briefly introduce what's coming next so students see the bigger picture:
- Introduction to CSS — how to style the HTML built today (colors, fonts, spacing)
 - Linking an external stylesheet with <link>

- The CSS box model — margin, border, padding, content
- Basic selectors — element, class, and id selectors

Tell students: "Today you built the skeleton. Tomorrow we give it skin and color."

24. Full List of HTML Entities Used Today (Reference Sheet)

Entity	Displays As	When To Use
<	<	Showing a tag as plain text
>	>	Showing a tag as plain text
&	&	Writing "and" symbol in text
"	"	Displaying quotation marks safely inside attributes
'	'	Displaying apostrophes safely
 	(non-breaking space)	Preventing browser from collapsing spaces
©	©	Copyright notices in footers
®	®	Registered trademark symbols
™	™	Trademark symbols
€	€	Currency values in Euros
Ź	₹	Currency values in Rupees

25. Teacher's Final Notes & Reflection Points

Before ending the class, walk through these reflection questions with students to consolidate learning:

- What is the difference between an element and a tag?
- Why can't we simply type the "<" symbol directly in visible page text?
- What's the risk of using multiple <h1> tags on one page?
- Why does the alt attribute matter even if the image loads perfectly for you?
- Name one real website feature that uses ordered lists and one that uses unordered lists.

Encourage students to answer out loud as a quick group recap — this cements the day's learning far better than silently re-reading notes.